

Predicting Myocardial Infarction Mortality

A clinically oriented evaluation of preprocessing, model selection, robustness, and final test-set performance.

PATIENTS

16,000~

UCI / Kaggle MI complications dataset

VARIABLES

124

Raw clinical columns before leakage control

FEATURES USED

104

After dropping leakage and other targets

FINAL MODELS

10

Baseline, tuned, ensembles, adversarial

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The clinical question: who is at risk before it is too late?

The goal is not only high accuracy; the goal is to identify high-risk patients without missing critical minority-class cases.

Problem

Post-MI patients can deteriorate through complications such as arrhythmia, shock, recurrent infarction, or death. Traditional scoring can miss non-linear interactions in high-dimensional clinical data.

Why ML?

Machine learning can combine demographics, ECG findings, blood pressure, laboratory values, medical history, and treatment indicators into a single risk model.

Why this is hard

The event is rare: the positive class is about 16%. A model can look accurate by predicting survival, while still missing the patients we care about most.

EHR data

Preprocess

Model

Evaluate

Clinical insight

Core evaluation principle: treat recall, F1-score, and ROC-AUC as more informative than accuracy alone.

Research objectives and evaluation strategy

A structured pipeline: from raw clinical data to final held-out test evaluation.

Objective	How it was tested
Predict mortality	Binary target LET_IS converted to survived vs died
Avoid leakage	Dropped ID, treatment-after-admission flags, and other complication targets
Handle clinical messiness	Missing value imputation, scaling, class imbalance strategies
Compare model families	LR, RF, XGB, SVM, KNN, NB, DT, voting, stacking
Measure robustness	15% label-noise injection and concept-drift simulation

What makes this presentation analytical

Every model decision is justified using exact numbers: Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix counts, threshold sensitivity, and feature importance.

Clinical priority

Recall is treated as a critical metric because false negatives represent missed deaths. F1-score balances recall with the false-alarm burden.

Academic priority

Final model choice is based on held-out test performance, not validation-only results, preventing optimistic bias.

High-dimensional myocardial infarction complication data

UCI/Kaggle dataset with clinical, ECG, laboratory, history, and outcome variables.

ROWS

16,000~

Patient admission records

RAW COLUMNS

124

Before target and leakage filtering

FEATURE MATRIX

104

Final predictors

POSITIVE RATE

16.1%

Validation/test mortality share

IMBALANCE

5.3:1

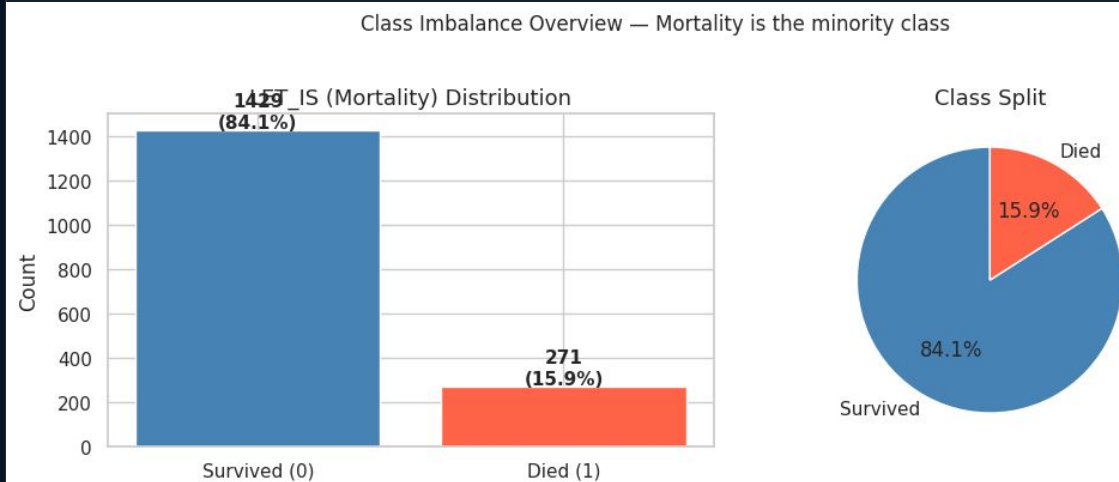
Survived to died ratio

Feature type	Count	Treatment
Continuous clinical/lab values	13	Median imputation + StandardScaler
Ordinal variables	3	Most-frequent imputation + StandardScaler
Binary flags	88	Mode imputation, no scaling
Dropped leakage/targets	19	Removed before training

Split	Shape	Positive rate
Train	1190 × 104	15.9%
Validation	255 × 104	16.1%
Test	255 × 104	16.1%

Target distribution reveals the main modeling risk

Mortality is the minority class, so accuracy alone is not enough.



IMBALANCE RATIO

5.3:1

Negative to positive class

POSITIVE RATE

~16%

Mortality/minority class

Interpretation

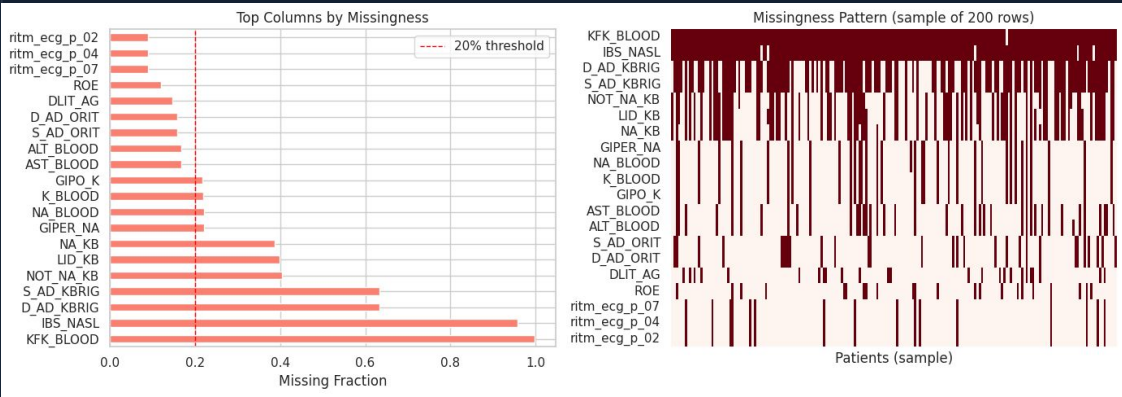
A naïve classifier could appear strong by predicting survival most of the time. The project therefore prioritizes recall and F1-score.

Clinical meaning

Every false negative is a patient who died but was predicted as low-risk. This is why threshold tuning is not a cosmetic step; it is central to deployment.

Data cleaning: missingness, outliers, and leakage control

Clinical datasets are valuable but noisy; preprocessing determines whether the model is trustworthy.

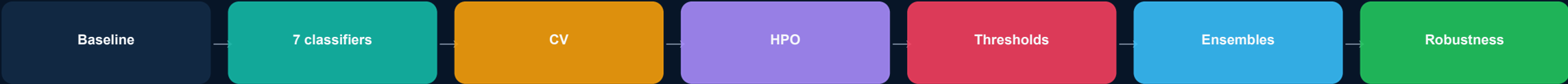


Data issue	Exact result	Action taken
Any missingness	103 columns	Type-aware imputation
High missingness	11 columns >20%	Tracked as quality risk
Overall missing rate	8.98%	Handled inside train-fit pipeline
Outliers	ROE 4.48%, L_BLOOD 3.56%, K_BLOOD 2.93%	Robust median imputation + scaling
Leakage risk	19 columns dropped	No post-outcome/treatment leakage



From baseline classifiers to robust final evaluation

The notebook compares simple, ensemble, tuned, thresholded, and adversarial-condition models.



Model family	Examples used
Linear models	Logistic Regression; baseline LR; tuned LR
Tree models	Decision Tree, Random Forest, XGBoost
Kernel / distance	SVM, KNN
Probabilistic	Naive Bayes
Ensembles	Soft Voting, Hard Voting, Stacking
Robustness models	Adversarial LR/RF trained on 15% noisy labels

Linear Regression checkpoint

The notebook target is binary mortality (LET_IS). Therefore classic Linear Regression and regression metrics such as R^2 , MAE, MSE, and RMSE are not the correct primary evaluation tools.

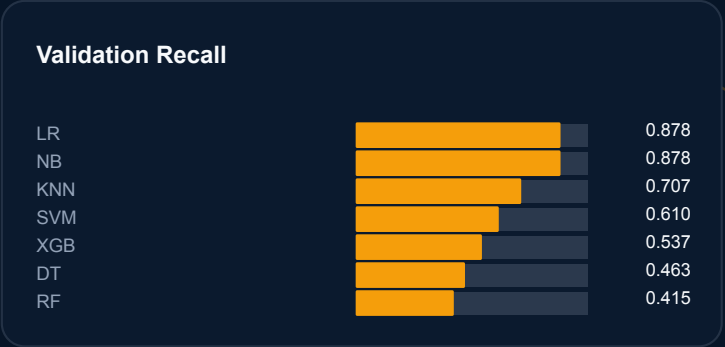
Correct alternative

Logistic Regression was used as the interpretable linear classifier, while Precision, Recall, F1-score, ROC-AUC, and confusion matrices were used for evaluation.

Initial validation results across 7 classifiers

The first full benchmark shows that model quality depends on the metric chosen.

Model	Acc.	Prec.	Recall	F1	AUC
XGBoost	0.8980	0.7586	0.5366	0.6286	0.9055
Logistic Regression	0.8078	0.4500	0.8780	0.5950	0.9053
Random Forest	0.8824	0.7391	0.4146	0.5312	0.8786
SVM	0.8196	0.4545	0.6098	0.5208	0.8464
Decision Tree	0.8588	0.5758	0.4634	0.5135	0.6765
KNN	0.7176	0.3258	0.7073	0.4462	0.7703
Naive Bayes	0.2039	0.1538	0.8780	0.2618	0.4826



Cross-validation and composite score prevent one-split overfitting

The top three selected models were Logistic Regression, XGBoost, and SVM.

Model	CV F1	CV AUC	Val Recall	Val F1
XGBoost	0.5498±0.0528	0.8808±0.0249	0.5366	0.6286
Logistic Regression	0.5419±0.0318	0.8550±0.0217	0.8780	0.5950
SVM	0.5458±0.0551	0.8469±0.0297	0.6098	0.5208
Random Forest	0.4160±0.0545	0.8696±0.0293	0.4146	0.5312
Decision Tree	0.4431±0.0508	0.6976±0.0479	0.4634	0.5135

Composite score formula

$$0.40 \times \text{Recall} + 0.35 \times \text{F1} + 0.25 \times \text{AUC} - 0.10 \times \text{CV_F1_Std}$$

Rank	Model	Composite
1	Logistic Regression	0.7826
2	XGBoost	0.6557
3	SVM	0.6323
4	KNN	0.6261
5	Random Forest	0.5660

Hyperparameter tuning: improvement is not guaranteed

Tuning exposed a key lesson: optimizing one objective can harm clinical recall.

Model	Recall before	F1 before	AUC before	Recall after	F1 after	AUC after
Logistic Regression	0.8780	0.5950	0.9053	0.8780	0.5950	0.8869
XGBoost	0.5366	0.6286	0.9055	0.5122	0.5915	0.9073
SVM	0.6098	0.5208	0.8464	0.1463	0.2264	0.8058

XGBoost

AUC slightly increased from 0.9055 to 0.9073, but F1 decreased from 0.6286 to 0.5915.

SVM

Recall collapsed from 0.6098 to 0.1463 after tuning, proving that HPO must match the clinical objective.

Takeaway

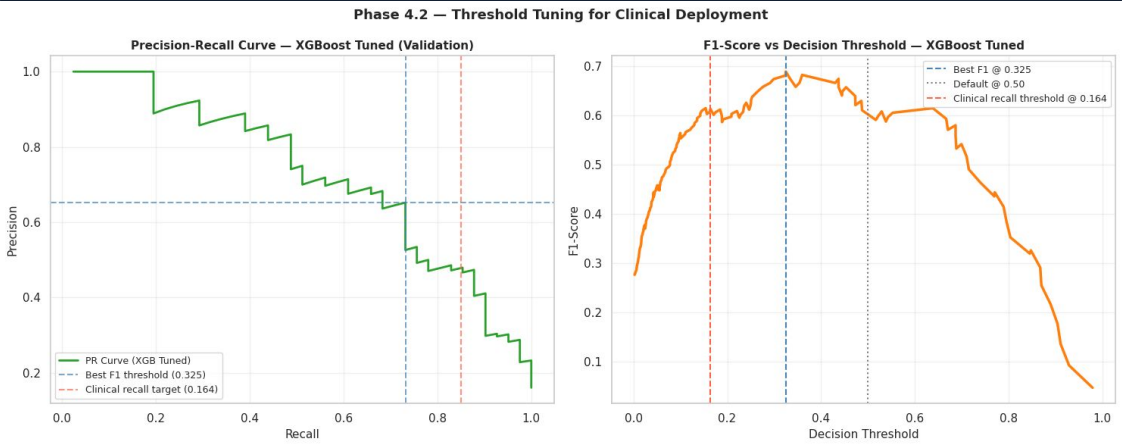
The project needs threshold tuning and recall-aware model selection, not only black-box hyperparameter search.

Class imbalance strategy and threshold tuning

The largest clinical gain came from changing the decision threshold, not just changing the algorithm.

XGBoost strategy	Recall	Precision	F1	AUC
SMOTE tuned	0.5122	0.7000	0.5915	0.9073
scale_pos_weight	0.6829	0.6667	0.6747	0.9070

Threshold strategy	Recall	Precision	F1	AUC
Default 0.50	0.3659	0.6818	0.4762	0.8668
Best F1 0.325	0.5610	0.5750	0.5679	0.8668
Clinical 0.164	0.7805	0.4267	0.5517	0.8668

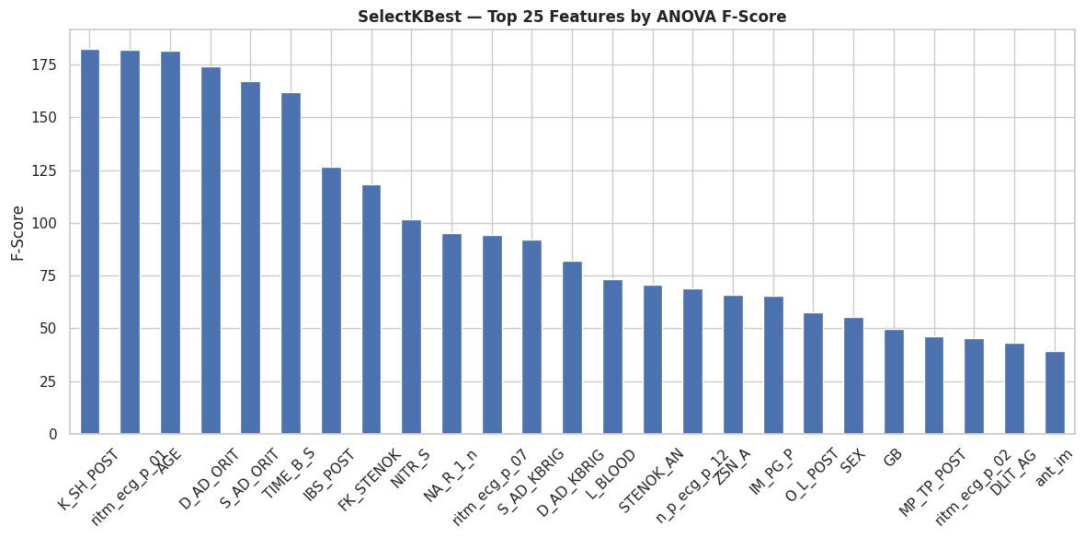


Feature importance and dimensionality reduction

The project compared full features, SelectKBest, consensus features, and PCA.

Feature set	N	Recall	F1	AUC	Precision
Full Features	104	0.5122	0.5915	0.9073	0.7000
Top-30 SelectKBest	30	0.5610	0.6389	0.8999	0.7419
Consensus	23	0.5610	0.6216	0.8898	0.6970
PCA-32	32	0.5610	0.5476	0.8319	0.5349

Top predictors by F-score	Score
K_SH_POST	182.33
ritm_ecg_p_01	182.03
AGE	181.35
D_AD_ORIT	174.07
S_AD_ORIT	167.17
TIME_B_S	162.01



Final held-out test set results: standard condition

The test set was locked until final reporting to avoid optimistic bias.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline LR	0.8314	0.4643	0.3171	0.3768	0.8043
Baseline RF	0.8627	0.8000	0.1951	0.3137	0.8576
Tuned LR	0.7608	0.3611	0.6341	0.4602	0.7960
Tuned XGB	0.8706	0.6818	0.3659	0.4762	0.8668
Tuned SVM	0.8667	1.0000	0.1707	0.2917	0.8132
Voting Soft	0.8667	0.7059	0.2927	0.4138	0.8525
Voting Hard	0.8863	0.7727	0.4146	0.5397	N/A
Stacking	0.8471	1.0000	0.0488	0.0930	0.8693
Adversarial LR	0.6157	0.2336	0.6098	0.3378	0.6801
Adversarial RF	0.8510	0.5600	0.3415	0.4242	0.8132

BEST F1 IN FINAL MODEL TABLE

0.5397

Voting Hard, Acc: 0.8863, Prec: 0.7727

BEST AUC

0.8693

Stacking, best F1 and 0.8693

BEST RECALL IN TABLE

0.6341

Tuned LR, misses 36.5% of deaths

Threshold-tuned XGBoost is even stronger for decision policy: Best-F1 threshold = 0.5679 F1; Clinical threshold = 0.7805 recall.

The model choice changes the clinical errors

Support on test set: 214 survived, 41 died. False negatives are missed mortalities.

Model / policy	TN	FP	FN	TP	Missed deaths	Interpretation
Tuned LR	168	46	15	26	36.6%	Highest recall among default models
Tuned XGB default	207	7	26	15	63.4%	High precision and AUC, weak default recall
Voting Hard	209	5	24	17	58.5%	Best F1 in final default-model table
XGB clinical threshold	170*	44*	9*	32*	22.0%	Best clinical recall policy; approximate counts from metrics

High recall

Tuned LR catches 26/41 deaths at default threshold; XGB clinical threshold catches about 32/41.

High precision

Voting Hard and Tuned XGB reduce false alarms, but they miss more deaths.

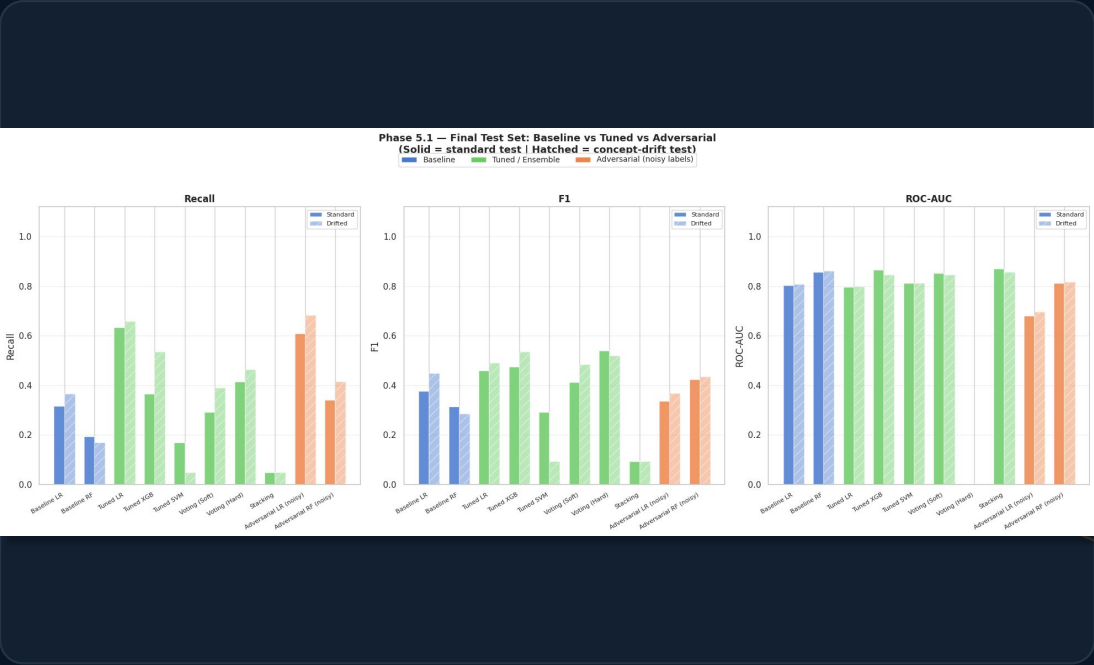
Clinical decision

For screening, accept lower precision to reduce missed mortalities. For reporting, show balanced F1/AUC.

Noisy labels and concept drift expose deployment risk

The final phase tests what happens when the world is messier than the training data.

Model	Acc.	Prec.	Recall	F1	AUC
Baseline LR	0.8549	0.5769	0.3659	0.4478	0.8082
Baseline RF	0.8627	0.8750	0.1707	0.2857	0.8607
Tuned LR	0.7804	0.3913	0.6585	0.4909	0.7990
Tuned XGB	0.8510	0.5366	0.5366	0.5366	0.8459
Voting Hard	0.8627	0.5938	0.4634	0.5205	N/A
Adversarial LR	0.6235	0.2523	0.6829	0.3684	0.6953
Adversarial RF	0.8275	0.4595	0.4146	0.4359	0.8165



Failure mode	Measured result
Best recall standard	0.6341
Best recall drifted	0.6829
Tuned LR F1 → noisy LR F1	0.4602 → 0.3378
Label-noise F1 drop	0.1224
Avg tuned-model F1 drop under drift	0.0093

Recommended model depends on the decision objective

For clinical use, the best model is not simply the highest accuracy model.

BALANCED ACADEMIC WINNER

XGB @ 0.325

F1 = 0.5679, Recall = 0.5610, Precision = 0.5750, AUC = 0.8668

CLINICAL SCREENING WINNER

XGB @ 0.164

Recall = 0.7805, F1 = 0.5517, Precision = 0.4267, AUC = 0.8668

DEFAULT-TABLE WINNER

Voting Hard

F1 = 0.5397, Accuracy = 0.8863, Precision = 0.7727

Why final recommendation is threshold-tuned XGBoost	Evidence
Best tunable clinical trade-off	Same AUC = 0.8668 while threshold controls recall vs precision
Best F1 policy observed	Best-F1 threshold improves F1 from 0.4762 to 0.5679
Clinically safest policy	Clinical threshold detects 78.05% of mortalities
More deployable than hard voting	Outputs probabilities, enabling calibrated risk thresholds

How the project aligns with medical ML research

The literature review supports the notebook’s emphasis on tree ensembles, imbalance handling, and interpretability.

Study theme	Reported benchmark	Relation to our project
Golovenkin et al. 2020	MI complications: F1 roughly 0.45–0.78 depending on outcome	Same high-dimensional MI complication domain; our XGB threshold F1 0.568 fits the realistic range
Soleimani et al. 2025	RF AUC = 0.924; XGBoost AUC = 0.905; LR-LASSO AUC = 0.893	Confirms tree ensembles are strong on tabular AMI data
Diakou et al. 2022	Same database; F1 around 0.54 and AUC <0.77 under standard preprocessing	Our thresholded XGB F1 0.568 and AUC 0.8668 are competitive
Zhang et al. 2025	LightGBM + SMOTE + Boruta + GSCV best; AUPRC emphasized	Supports future work: LightGBM, stronger feature selection, and PR-focused metrics
Lee et al. 2021	ML outperformed TIMI/GRACE for NSTEMI long-term mortality, AUC up to 0.840	Our AUC results are in a clinically plausible range but still need external validation

What failed, and why it matters

The project demonstrates technical progress while remaining honest about clinical deployment limits.

Persistent recall gap

Even the best standard recall is 0.6341, meaning 36.6% of mortalities are still missed at default policy.

SVM tuning pitfall

Tuned SVM recall collapsed from 0.6098 to 0.1463, showing that HPO can optimize the wrong behavior.

Label-noise amplification

15% label flips reduced LR F1 from 0.4602 to 0.3378; SMOTE can amplify corrupted minority examples.

Missingness risk

103 feature columns contained missing values and 11 had more than 20% missingness.

No external validation

The model is tested internally only; hospital-to-hospital generalization remains unproven.

Interpretability gap

High-performing models need SHAP/LIME explanations before clinical trust is realistic.

Key maturity statement: this is a strong academic ML pipeline, not yet a deployable clinical device.

How to turn the notebook into a stronger research pipeline

The next iteration should focus on reliability, clinical alignment, and interpretability.

Priority	Recommendation	Expected impact
1	Temporal holdout instead of random split	More realistic deployment estimate and drift detection
2	Recall-optimized HPO using recall or F β scoring	Prevents SVM-style recall collapse and aligns with clinical risk
3	Cleanlab/confident learning before SMOTE	Detects mislabeled examples before synthetic oversampling amplifies noise
4	LightGBM/CatBoost + calibrated probabilities	Stronger tabular performance and more reliable thresholding
5	SHAP explanations for top predictors	Improves clinical trust and detects impossible feature behavior
6	AUPRC and calibration curves	Better evaluation under severe class imbalance than ROC-AUC alone
7	External validation across hospitals	Tests generalizability and reduces single-dataset bias

A strong ML pipeline with a clinically realistic model decision

The best answer is not “one model wins”; it is “choose the model and threshold for the clinical objective.”

MAIN RESULT

XGB + threshold

Best-F1 threshold: F1 0.5679; Clinical threshold: Recall 0.7805

MOST IMPORTANT INSIGHT

Accuracy is misleading

Best default accuracy 0.8863 still misses 58.5% of deaths in Voting Hard

DEPLOYMENT CAVEAT

Validate first

Needs temporal split, external validation, calibration, and explainability

Final message

The project successfully moves from raw MI data to a defensible, numerically justified final model recommendation: threshold-tuned XGBoost for clinical screening, supported by rigorous EDA, leakage control, cross-validation, feature selection, robustness testing, and honest limitation analysis.

Thank you

PREDICTING MYOCARDIAL INFARCTION MORTALITY

A clinically oriented evaluation of preprocessing, model selection, robustness, and final test-set performance.

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1. The Clinical Problem

- **Post-MI patients** deteriorate through fatal complications
- **Traditional** scoring misses non-linear interactions
- **Challenge:** Rare positive mortality class (~16%)

- A model can look highly accurate just by predicting survival, missing high-risk patients



2. Dataset & Preprocessing

- **Source:** UCI / Kaggle MI complications dataset 
- **Records:** 1,700 patient admissions
- **Features:** 104 clinical predictors
- **Handling 8.98% Missing Data** (Median & Mode Imputation within pipeline)



3. Methodology & Evaluation Strategy

- Priority:** Recall, F1-score, and ROC-AUC over Accuracy
- Algorithms → LR, RF, XGB, SVM → KNN, NB → Stacking Ensemble
- **Clinical Priority:** Every false negative represents a missed death
 - **Models Tested:** LR, RF, XGB, SVM, KNN, NB, DT, Voting & Stacking Ensembles
 - **Robustness Tests:** 15% Label-Noise Injection & Concept Drift Simulation



4. Key Results & Model Selection

- Best baseline accuracy 0.8863 (missed 58.5% deaths)
- **Threshold Tuning is Crucial** for gains

FOCUS

SCREENING WINNER:
XGBoost
(Threshold 0.164)
Recall: **78.05%**
F1: **0.5517**
AUC: **0.8668**

FOCUS

BALANCED WINNER:
XGBoost
(Threshold 0.325)
F1 Score: **0.5679**
Recall: **56.10%**
AUC: **0.8668**



5. Honest Limitations

- **Persistent Gap:** Best baseline misses 36.6% mortalities
- **Tuning Pitfalls:** HPO can optimize wrong behavior
- **No External Validation:** Hospital-to-hospital generalization is unproven 



6. Recommendations

- **External Validation:** Temporal holdout & cross-hospital testing 
- **Recall-Optimized HPO:** Align with clinical risk
- **SHAP Explanations:** Improve clinical trust & interpretability